Computer Networks
Observation

INDEX

NAME : Dhana Sri. S STD : ..... SEC : ..... ROLL NO : .....

SUBJECT : ..... SCHOOL : .....

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Completed
M. Dhanasri 12/12/25

In Case of Emergency Please Contact Mobile :

20/1/26

Exercise - 1

Aim:-
Study of Various Network commands used in Linux and Windows.

1 arp -a:-
Description:- It shows the Ip address & MAC address of the computer
O/p:-
gateway (10.0.2.2) at 52:55:0a:00:02:02 [ether]

2. hostname
Des:- Displays the name of the Computer
O/p:-
fedora

3. Ipconfig/all
Des:- Detailed information about Tcp/IP connection including Router, Gateways, DNS, DHCP. type of Ethernet Adapter.

O/p:-
Primary WINS Server 192.168.253.2
NetBIOS Over Tcpip Enabled

4 nbtstat -a
Des:- Solve the problem with NetBIOS name resolution. (Nbt stands for Net-BIOS over TCP/IP)

O/p:-
RemoteName Remote host machine name
Ip address Dotted decimal representation of Ip addre
Interval Redisplays selected statistics, pausing Interval second

5 netstat
Des:- Displays a variety of statistics about a computers active Tcp/Ip condition.

O/p:-

Active Connections

Proto	Local Address	Foreign Address	State
TCP	172.16.10.182:7680	DESKTOP-LUKJJ8:60548	Established

6. nslookup

Des- (name server lookup) displays a variety of statistics about a computer active TCP/IP connection.

O/p:- Default Server: Unknown

Address: 172.16.8.1

7. nslookup www.google.com

Des:- Displays the server & address

O/p:-

Server: www.google.com

Addresses: 2001:4860:4827:7100::

⋮

No response from server

8. Pathping

Des: It is unique to Windows & it is basically a combination of ping & Tracert commands.

O/p:-

options:

-g host list loose source route along host-list

9. ping www.facebook.com

Des: (Packet INternet Grouper) command to test Connectivity btw two Nodes.

O/p.

Reply from 57.144.56.1: bytes=32 time=4ms TTL=56

Minimum = 4ms, Maximum = 5ms, Avg = 4ms

* ping localhost

o/p :- pinging HDC042257 [::1] with 32 bytes of Data

* ping 4.2.2.2

o/p: Packets: Send = 4, Received = 4, Lost = 0 (0% Loss)
Approximate round trip times in milli-seconds.

11. Route cmd

Dep:- It is used to show the Ip routing table. It is primarily used to setup.

o/p:-

route PRINT
route print -4
route DELETE 157.0.0.0
route DELETE 3ffe::/32

IP routing table

Destination	Gateway	Genmask	Flags	Metric	Ref	Se Iface
default	gateway	0.0.0.0	UG	100	0	

Some Important Linux Commands

1. Ip :-

The ip cmd is basic cmds every administrator will need in work. It shows address Info, manipulate routings, display network various devices, Interfaces, & tunnels.

Eg: ip address show

o/p:-

1: lo: <LOOPBACK, UP, LOWER_UP> mtu
65536 qdisc noqueue state UNKNOWN
group default qlen 1000
valid_lft forever preferred_lft forever

2. ifconfig

Command in many sysadmin's tool belt for configuring & Troubleshooting network

o/p:

```
enp2s0: flags=4163<UP,BROADCAST,RUNNING,
  MULTICAST>  mtu1500
  inet 172.16.9.200  netmask 255.255.252.0
  broadcast 172.16.11.255
```

3. mtr

Matt's Traceroute is a network diagnostic tool that combines the feature of ping & Traceroute. It displays the route to a destination while continuously measuring latency & packet loss at each hop.

o/p:-

```
fedora (::1)  -  localhost (::1)
Keys: Help Display mode Restard Statistics Order
of fields quit
Host                    Packet
1. local host           Loss%  Snt  Last
                        0.0%   83   0.1
```

mtr -g google.com

o/p: -n, --no dns do not resolve host names
-h, --help display this help & exit
-v, --version output version information & exit

4. tcpdump -D

It is designed for capturing & displaying packets

You can install tcpdump with cmd below

[root@server~] # dnf install -y tcpdump

O/P:-

1. enp0s3 [UP, Running, Connected]
2. any [pseudo device that captures on all interfaces]
3. lo
4. bluetooth-monitor [wireless]
5. usbmon2

5. Ping google.com

Ping is a tool that verifies IP level connectivity to another TCP/IP computer by sending ICMP echo request messages. The receipt of corresponding echo reply msg is displayed.

O/P:- (ctrl+c to stop)

64 bytes from pnmaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=159 ttl=255 tm

Configuring an Ethernet connection by using nmcli

IP commands

1) a:

IP address add 192.168.1.254/24 dev enp0s3

Assigns the IP address 192.168.1.254 with subnet mask /24 to interface enp0s3.

O/P:- command executed successfully

cmd: IP address show enp0s3 (To verify)

O/P:- 3: enp0s3: <BROADCAST, MULTICAST, UP, LOWER_UP>
inet 192.168.1.254/24 scope global enp0s3

2) IP address del 192.168.1.254/24 dev enp0s3

It removes the specified IP address from the Interface

cmd: ifconfig address show enp503 (verify)

op:

3: enp503: <BROADCAST, MULTICAST, UP, LOWER_UP

3) ip link set eth0 up

It Activates the Network Interface

ip link show eth0 (verify)

O/p:

2: eth0 <BROADCAST, MULTICAST, UP, LOWERUP> mtu 1500 mtu 1500 qdisc fq_codel state up group default

4) ip link set eth0 down

It disables the network Interface

ip link show enpos3

enpos3: <BROADCAST, MULTICAST> mtu 1500 qdisc fq_codel DOWN mode DEFAULT group default qlen 100

link/ether 08:00:27:02:df:se brd ff:ff:ff:ff:ff:ff

5) ip link set enPO33 promise on ip link show

enpos3: <BROADCAST, MULTICAST, PROMISC> mtu 1500 qdisc fq_code state DOWN DEFAULT qlen 1000

link/ether 08:00:27:02:df:8e brd ff:ff:ff:ff:ff:ff

6) ip route add default via 192.168.1.254 dev enp0s3

7) ip route get
10.10.1.14 via 10.0.2.2 dev enp0s3 src 10.0.2.15 id 1000

Configuraring are Ethernet Connection by using nmcli
If you connect a host to the network over Ethernet you can manage the connection setting

1) nmcli connection show
NAME UUID TYE DEVICE
Wired connections dfd1365f-0d9-3c63 ethernet etho
lo 34b0b00e-983e-4941-a00b loopback lo
bcdes87f8a38

2) nmcli connection add type ethernet if name etho
Con-name "wired connection 1"
Connection 'wired connection" (a00c99-7513-4e63-92ee-40edd0f4adc) successfully added

nmcli connection show
NAME UUID TYPE DEVICE
wired connection dfdf35f.c2d9-3eb3-ab4f ethernet e
lo 34b00e-983e-4941-a0ab loopb
wired connection a00c999-7513-4c63-92ee-40edd0
c4adc ethernet

3) ip address show etho
etho: <BROADCAST, MULTICAST, UP, LOWER UP> mtu 1500 qdisc fq
link state UP group default qlen 1000
link/ether 08:00:27:8a:35:d2 brd ff:ff:ff:ff:ff:ff
inet 192.0.2.1/24 brd 192.0.0.255 scope global noprefixroute

Student Observation

1. Which command is used to find the reachability of a host machine from your device.

Sol The ping command is used to check whether a host machine is reachable from your device.

2. Which cmd gives the details of hops taken by a Packet to reach its destination.?

Sol The mtr cmd shows the route from a computer to a specified host and provides information about each hop.

3. Which command display IP Configuration of your machine?

Pd address show

It display IP address assigned to network Interface.

4. Which cmd displays ~~information~~. Tcp port Status on your machine?

net stat

It displays information about achieve TCP/IP connection & network statistics

5. Write the cmd to modify the IP Configuration in a linux machine

To assign an IP address

ip address add 192.168.1.254/24 dev enp53

To delete an ip address

ip address del 192.168.1.254/24 dev enp053

Ex.No:2
28/7/26

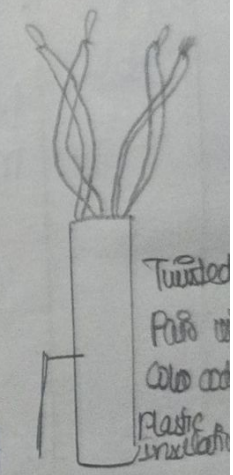
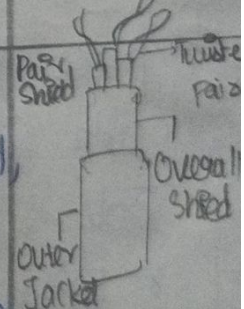
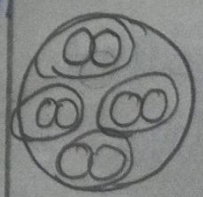
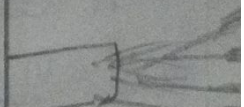
Ex: 2

Aim: Study of different types of Network Cables.

a) Understand different types of network cable

Different type of cables used in networking are:

1. Unshielded Twisted Pair (UTP) Cable.
2. Shielded Twisted Pair (STP) Cable
3. Coaxial Cable
4. Fibre Optic Cable

Cable Type	Category	Maximum Data transmission	Advantages / Disadvantages	Application / Use	Image
UTP	Category 3	10 bps	Adv: * Cheaper in cost * Easy to install as they have a smaller overall Diameter Dis: * More prone to (EM) electromagnetic Interference & noise	10Base-T Ethernet Fast ethernet Gigabit ethernet Fast Ethernet Gigabit Ethernet	
	Category 5	Up to 100 mbps			
	Category 5e	1Gbps			
STP	Category 6, 6a	10Gbps	Adv * Shielded * Faster than UTP * Less Susceptible to noise & Interference Dis * Expensive * Complex Installation effort	Gigabit Ethernet, 10G Ethernet (55m) widely used in data centres Gigabit Ethernet, 10G Ethernet (100 m)	  
SSTP	Category 7	10Gbps			

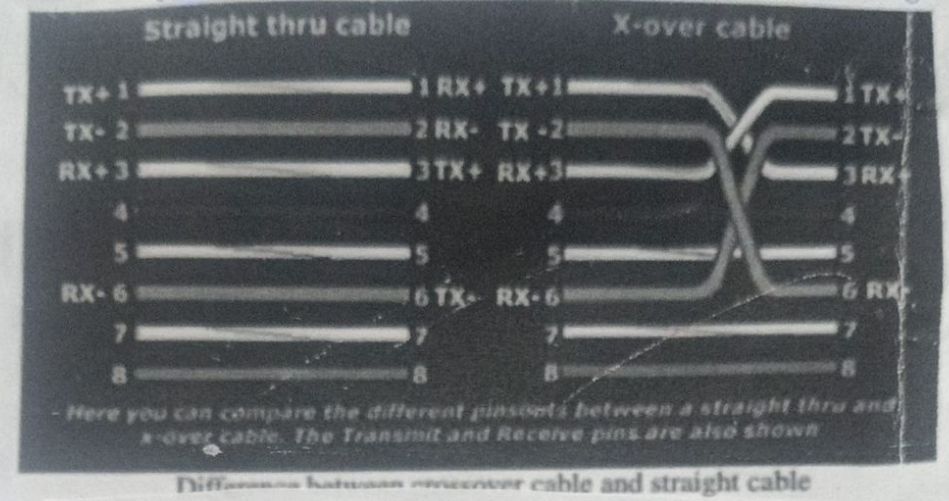
Coaxial Cable	RG-6 RG-59 RG-11	10-100 Mbps	* High Bandwidth * Immune to Interference * Low loss Bandwidth * Versatile Dis * Limited distance * Cost * Size Is Bulky	Speed of signal is soom Television network High speed internet connection	Braided Shield Outer Jacket
Fibre optics cable	Single Mode Multi mode	100 Gbps	Adv: * High speed * High Bandwidth * High security Disadv * Expensive * Requires skilled Installers	* Maximum distance of fibre optics cable is around 100 meters	

b) Make your own Ethernet Cross-Over Cable / Straight Cable

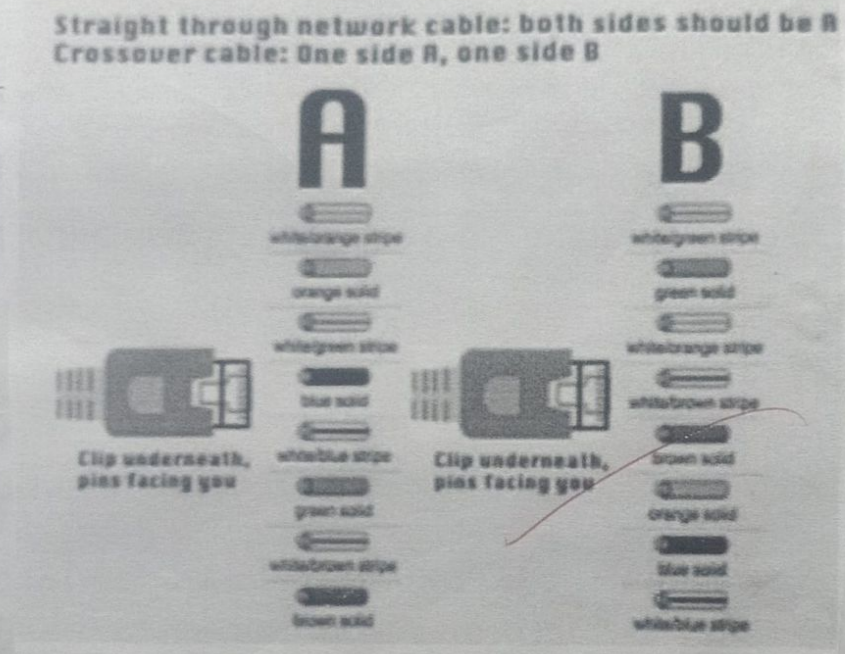
Tools & parts needed:

* Ethernet Cabling. CAT5e is certified for gigabit support, but CAT5 cabling works as well, just over short distances
* A crimping tool. This is an all-in-one network tool shaped to push down the pins on the plug & strip & cut the shielding of cables

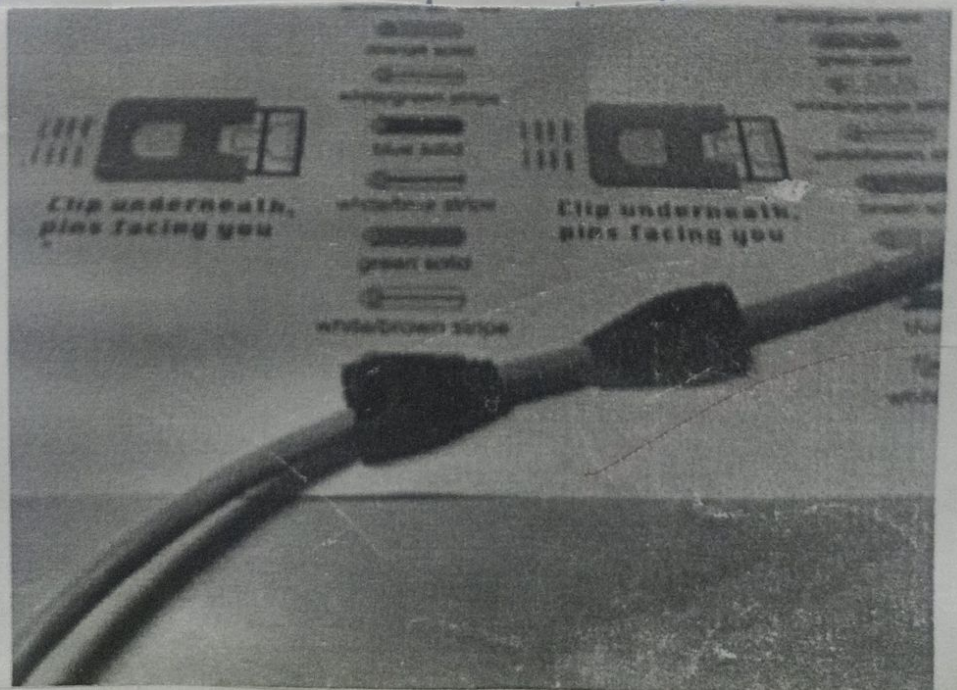
* Two RJ45 plugs
* Optional two plugs shields
Diff btw Crossover cable & Straight cable



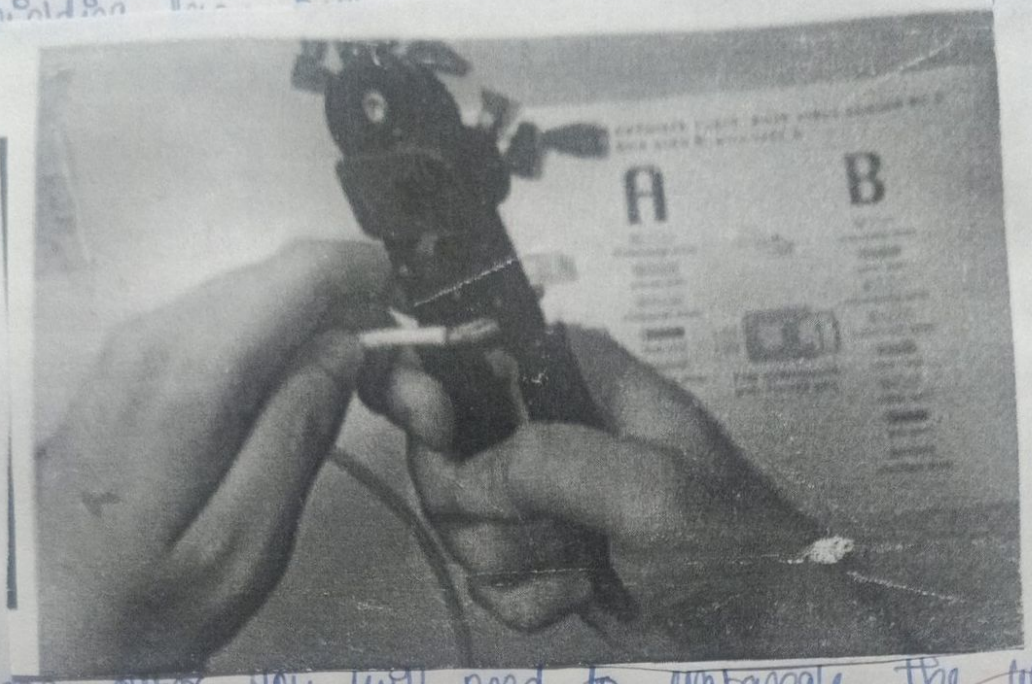
Difference between crossover cable and straight cable



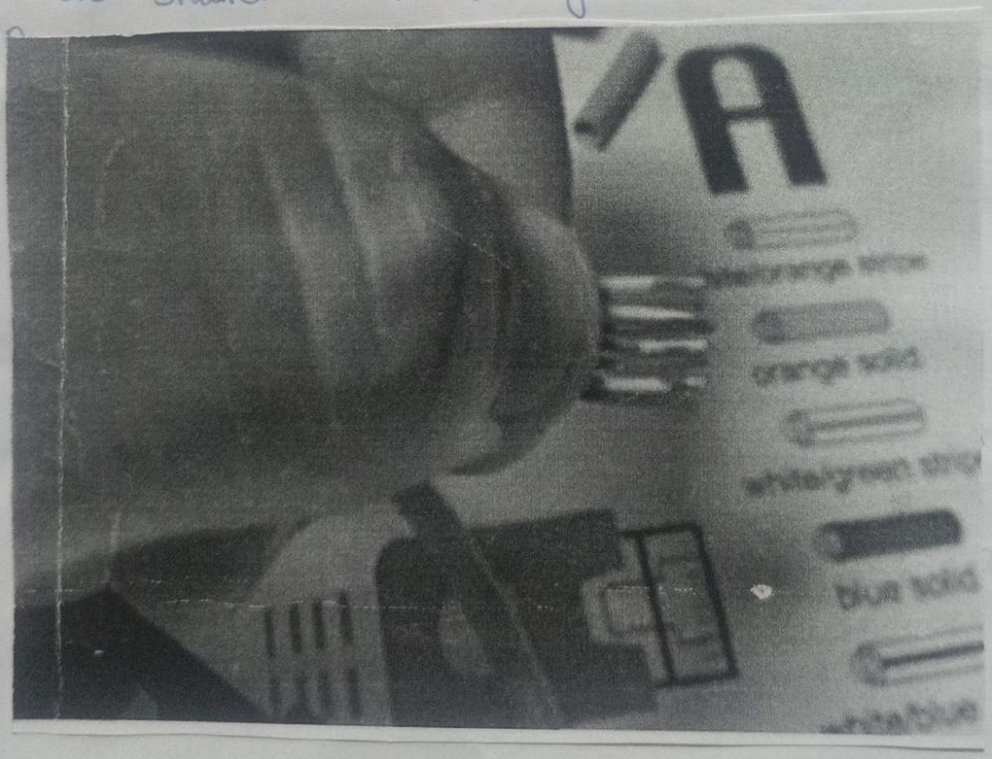
Step 1: To start construction of the device, begin by



Step 2: Next Strip approximately 1.5 cm of cable sheathing. Inside you will find 4 pairs of wires, each has a

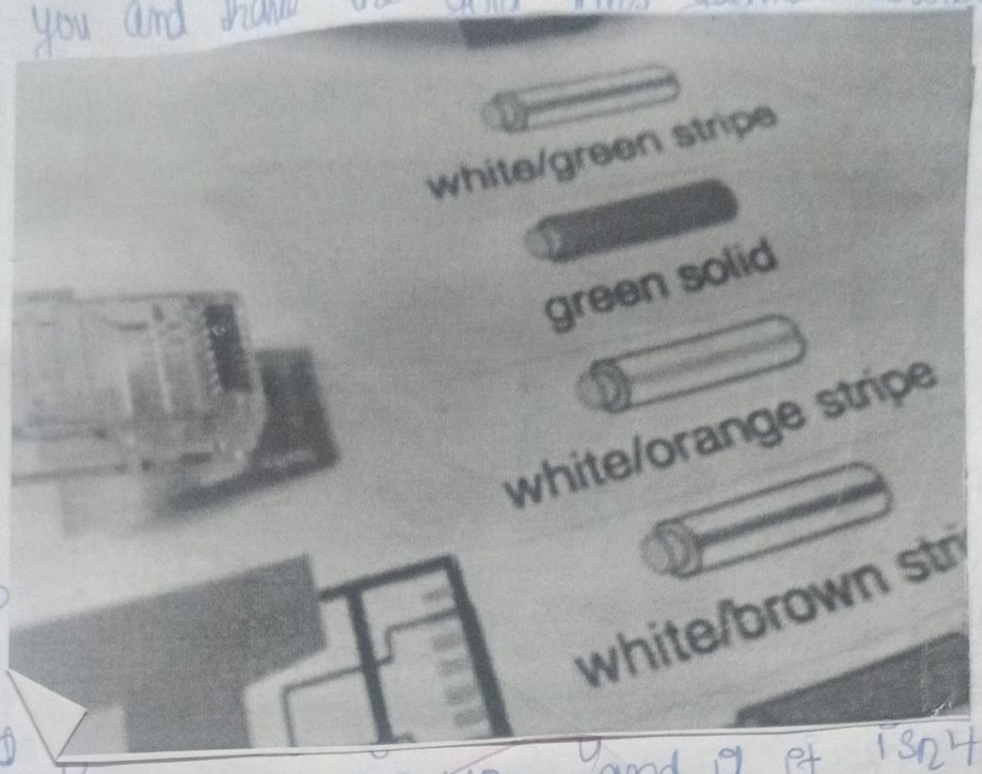


Step 3: After you will need to untangle the wires; there should be four "twisted pairs". Referencing back to the sheet, arrange them from top to bottom. One end should be in arrangement A and the other in

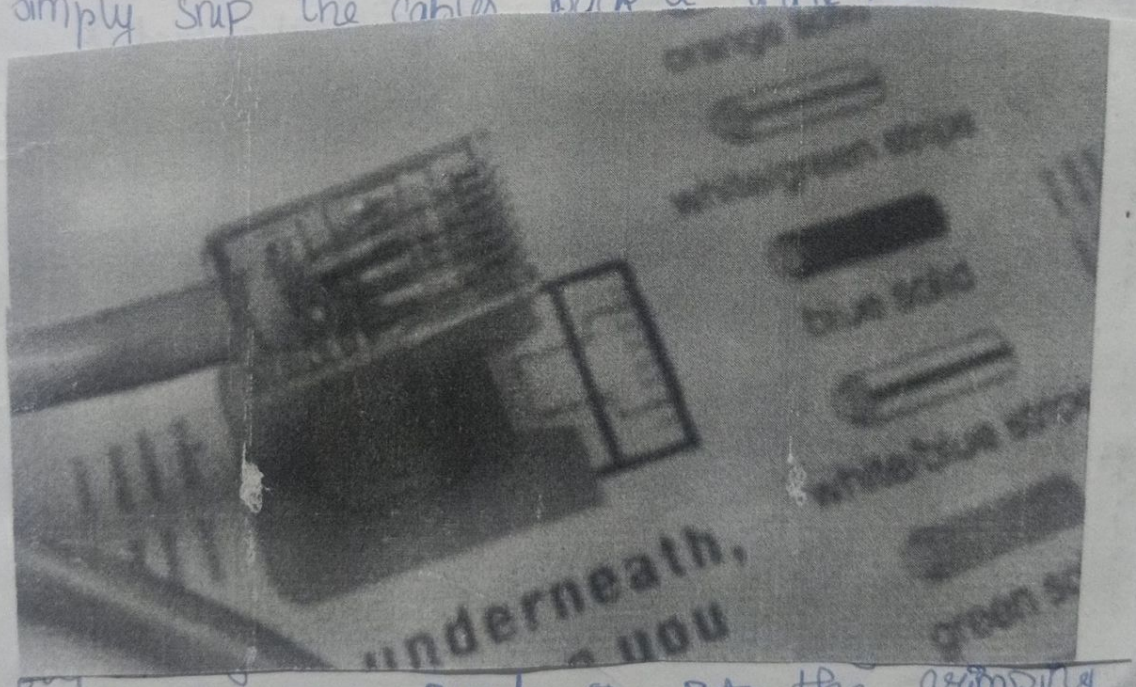


Step 4: Once the order is correct, bunch them together in a line, and if there are any that stick out further than others, snip them back to create an even level. The difficult aspect placing these into RJ45 plug without messing up the order. To

do so, hold the plug with the clip side facing away from you and have the gold pins facing towards you,



Step the just end over the cable shielding, and if it isn't that means that you stripped off too much shielding. Simply snip the cables back a little more



Inside the plug, insert it into the crimping tool and push down.

Step 7: Lastly, repeat for other using diagram B (to make a crossover cables) / using diagram A (to make a straight through cable)

~~Take~~ To take it, plug it in and attempt to

Connect two devices directly

Student observation:

1. What is the difference between cross cable and straight cable?

Sol: Straight cable has the same wiring on both ends and connects difference devices. Cross cable has different wiring on each and connects similar devices.

2. Which types of cable is used to connect two PCs?

Sol Cross Cable.

3. Which type of cable is used to connect a router/switch to your PC?

Sol Straight cable.

4. Find out the category of twisted pair cable used in your lab to connect the PC to the network socket

Sol Cat 6 cable

5 Write down your understanding, challenges faced, & output received while making a twisted pair cross/straight cable.

Sol Understanding: Learned to make & test straight/cross tables

Challenges:- Arranging wires correctly & crimping the RJ-45 connectors

Ex.No:3
04-08-26

Ex-3

Aim: To Study the packet Tracer tool Installation & User Interface Overview

② To understand environment of CISCO PACKET TRACER to design simple network.

Introduction:

1- It allows you to model complex systems without the need for dedicated & its environment.

2. It helps you to practice your network configuration & troubleshooting skills via computer or an Android.

3. It is available for both Linux & window desktop Environment

Installing packet Tracer:

Windows:

Installation in Windows is pretty simple & straight forward the setup comes in a single. file named Pack.tracer - Setup 8.0.1 . exe.

Linux:

Linux users with an Ubuntu distribution should download the file for Ubuntu, & those using Fedora:

· /packet Tracer601_i386_Installer-rpm-bin

User Interface Overview:-

The Layout of packet Tracer is divided into Several Components.

1. Menu bar - This is common menu found in all software.
2. Main toolbar - This Bar provides short cut keys to menu.
3. Logical/physical workspace tabs - These tabs allow you to toggle btw logical & physical work area
4. Workspace
5. Common tools Bars - This toolbar provides control for manipulating topologies.
6. Real-time/simulation tabs
7. Network Component box
8. User-created packet Box

2) Analyse the Behaviour of network devices using Cisco Packet Tracer Simulator.

1. From Network Component Box, Click & drag & drop below Components.
   a. 4 Generic PC & one HUB
   b. 4 Generic PCs & one Switch
2. Click on Connections
   a. Click on Copper Straight - Through cable
   b. Select one of PC & connect it to HUB using cable
   c. Similarly connect 4 PCs to the Switch using copper straight - through cable.
3. Click on PCs connected to hub, go to the Desktop tab, Click on IP Configuration.

4. Observe flow of PDU from source PC to destination PC by selecting Real Time.

5. Repeat step 3 & 5 for the PCs connected to the switch.

6. Observe how HUB & switch are forwarding & PDU & write your observation & conclusion about the behaviour of switch & HUB

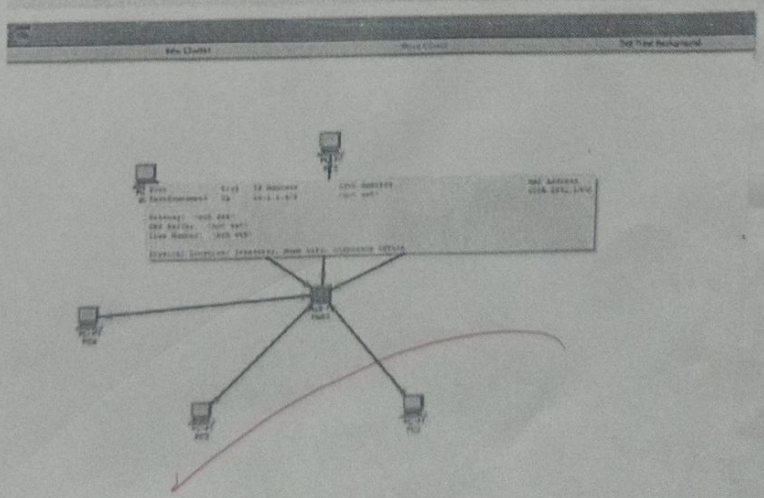
Student Observation

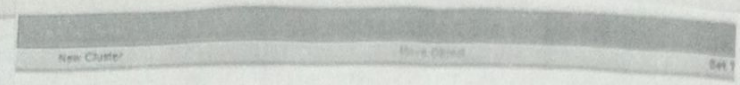
a. From you observation, write down the behavior of switch & HUB in terms of forwarding the packets received By them.

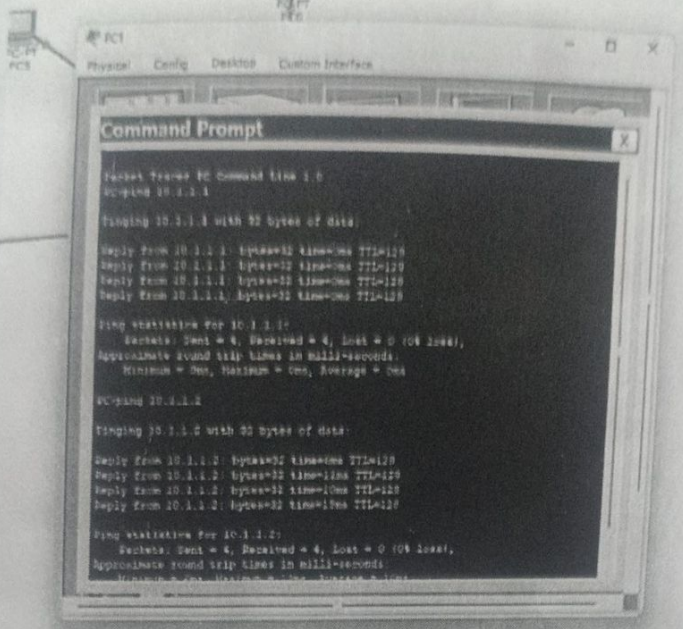
A. Hub: Forwards data to all connected devices.
Switch: Forwards data only to destination device using its MAC address

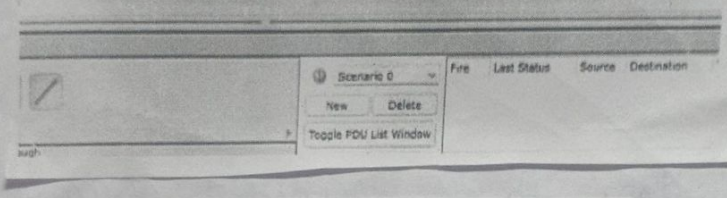
b. Find out network topology implemented in your college & draw and label that topology.

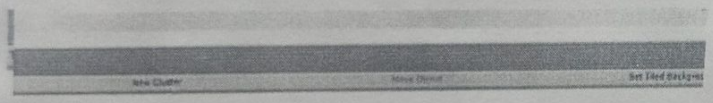
A Star Topolog

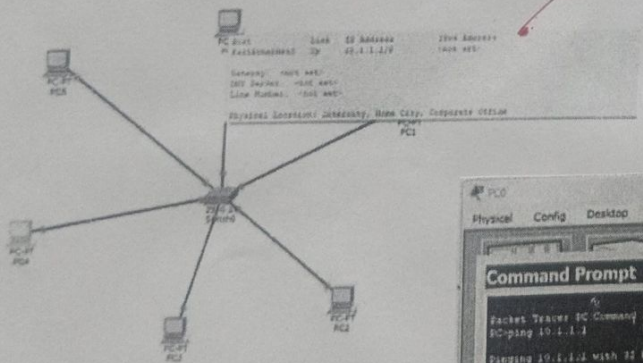


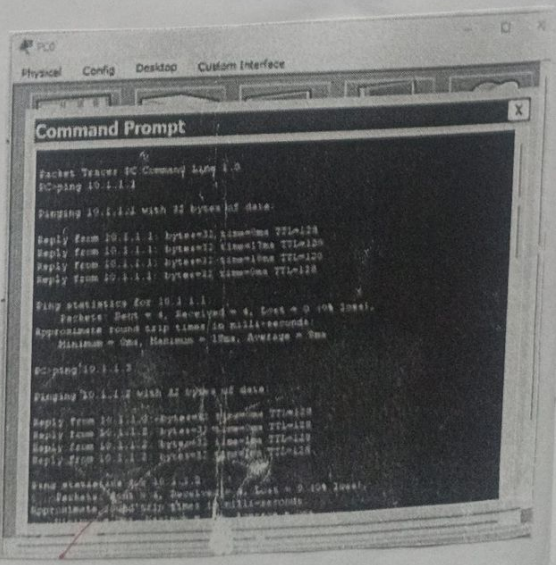



PC1
Physical
Config
Desktop
Custom Interface
Command Prompt
Packet Tracer PC Command Line 1.0
PC>ping 10.1.1.1
Pinging 10.1.1.1 with 32 bytes of data:
Reply from 10.1.1.1: bytes=32 time=0ms TTL=128
Reply from 10.1.1.1: bytes=32 time=0ms TTL=128
Reply from 10.1.1.1: bytes=32 time=0ms TTL=128
Reply from 10.1.1.1: bytes=32 time=0ms TTL=128
Ping statistics for 10.1.1.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
PC>ping 10.1.1.2
Pinging 10.1.1.2 with 32 bytes of data:
Reply from 10.1.1.2: bytes=32 time=0ms TTL=128
Reply from 10.1.1.2: bytes=32 time=12ms TTL=128
Reply from 10.1.1.2: bytes=32 time=10ms TTL=128
Reply from 10.1.1.2: bytes=32 time=19ms TTL=128
Ping statistics for 10.1.1.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:


Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination






PC0
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PC>ping 10.1.1.1
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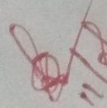
Reply from 10.1.1.1: bytes=32 time=1ms TTL=128
Reply from 10.1.1.1: bytes=32 time=10ms TTL=128
Reply from 10.1.1.1: bytes=32 time=0ms TTL=128
Ping statistics for 10.1.1.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 10ms, Average = 0ms
PC>ping 10.1.1.2



Ex:4
10-08-26

Ex:-4

Aim: Setup and configure an LAN using a Switch & Ethernet Cables in your Lab.

~~Equipment Required~~ To design, Set up, & configure a LAN connecting three computers through a network Switch, assign Static Ip addresses to all devices, Secure verify Connectivity using ping and & demonstrate resource / data Sharing across the network

2. Equipment Required :-	Quantity	Specification
1. personal Computer (pc)	3	Ethernet (NIC) enabled
2. Network Switch	1	8/16/24 port, unmanaged/managed
3. Ethernet Cables (cat5e/ Cat 6)	4	Straight - through, RJ-45 connectors
4. Server	1	

3. Theory

A LAN is a network that connects computers and devices within a limited geographical area such as home, office, or Laboratory allowing them to share resources such as files, printers & Internet Connections.

A star Topology is used in this experiment, where all end devices (PCs) are individually connected to Central network switch. Every device on a LAN must be assigned an Ip address within same network range.

4. Network diagram

The LAN was implemented using star Topology as show as below.

5. Ip address Configuration Table

Devices	Ip address	Subnet Mask	Connected To
PC0	10.1.1.1	255.0.0.0	Switch Port Fa0/1
PC1	10.1.1.2	255.0.0.0	Switch Port Fa0/2
PC2	10.1.1.3	255.0.0.0	Switch port Fa0/3
PC3	10.1.1.4	255.0.0.0	Switch Port Fa0/4
Switch	10.1.1.5	255.0.0.0	VLAN 1 Interface

6. Implementation in Cisco Packet Tracer

Since physical hardware was limited, the LAN was also implemented & verified.

Step1: Place the devices

1. Open Cisco Packet Tracer.
2. From the device category panel, select end devices & drag 4 PCs onto the work space.
3. Select Switches & drag 1 Switch onto the workspa

Step 2: Cable the devices

4. Click Connections → Select Copper Straight-Through Cable.
5. Click each PC, then click switch, & select an available Fast Ethernet port to complete each connections.
6. Repeat for all four PCs. Green link lights confirm a successful physical connections.

Step 3: Assign Ip addresses to PCs.

7. Click a PC → Desktop tab → IP Configuration
8. Enter the Static Ip address and Subnet Mask as per Ip table for each PC.

Step 4: Configure the Switch Management IP

9. Click the switch → CLI tab.

10. Enter the following cmds to assign a management IP:

Enable
Configure terminal
Interface vlan 1
Ip address 10.1.1.5 255.0.0.0
No shutdown
Exit

Step 5: Verify Connectivity using Ping

11. Click a PC → Desk Top tab → Cmd prompt

12. Use the ping cmd to test connectivity to the other PC. eg: ping 10.1.1.2

13. Successful "Reply from..." messages confirm connectivity

Step 6: Demonstrate Data sharing (Server Simulation)

14. Add a server device (End devices → Server-PT) & connect it to the switch.

15. Assign PT an IP address (eg: 10.1.1.10, subnet mask 255.0.0.0) via its Desktop → IP Configuration tab.

16. Open the Server's Services (config) tab & enable the HTTP service.

17. From another PC, open the Web Browser & enter the Server's IP address to load the page, confirming successful data Transfer across LAN1.

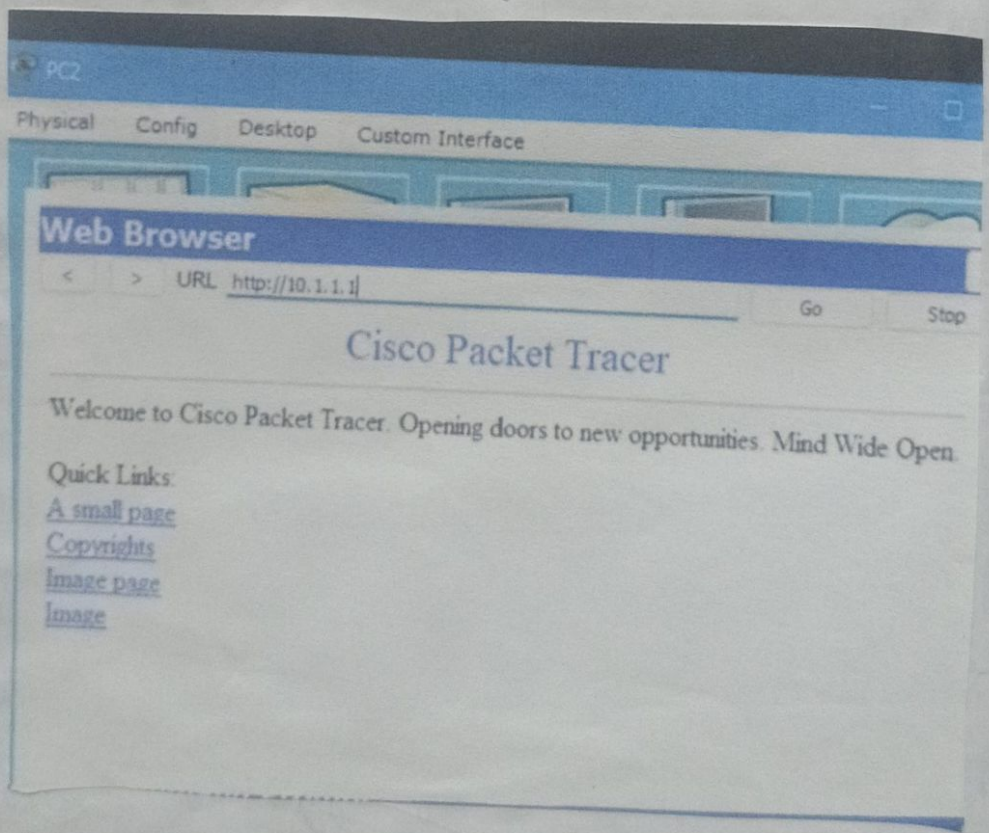
Step 7: Observe Packet Flow in Simulation Mode

18. Switch to Simulation mode (bottom-right of the workspace).

19. Send a ping or web request between two devices & use the play/capture controls to step through packet as it travels hop-by-hop through switch, observing MAC-address-based forwarding.

9. Observation

All four PCs were successfully connected to the switch in a star topology & assigned unique static IP addresses within the same subnet. Ping tests between all devices returned successful replies, confirming end-to-end connectivity & functioning correctly for resource sharing.



```
Packet Tracer PC Command Line 1.0
PC>ping 10.1.1.1

Pinging 10.1.1.1 with 32 bytes of data:

Reply from 10.1.1.1: bytes=32 time=0ms TTL=128
Reply from 10.1.1.1: bytes=32 time=0ms TTL=128
Reply from 10.1.1.1: bytes=32 time=0ms TTL=128
Reply from 10.1.1.1: bytes=32 time=0ms TTL=128

Ping statistics for 10.1.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

PC>ping 10.1.1.2

Pinging 10.1.1.2 with 32 bytes of data:

Reply from 10.1.1.2: bytes=32 time=1ms TTL=128
Reply from 10.1.1.2: bytes=32 time=0ms TTL=128
Reply from 10.1.1.2: bytes=32 time=6ms TTL=128
Reply from 10.1.1.2: bytes=32 time=9ms TTL=128

Ping statistics for 10.1.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
```

Physical Config CLI

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to
up

%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to
up

%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to
up


Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface vlan 1
Switch(config-if)#ip address 10.1.1.5.255.0.0.0
                             ^
% Invalid input detected at '^' marker.

Switch(config-if)#no shutdown

Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up
exit
Switch(config)#
```

Copy Paste

15/9/26
EX: 5

Ex:- 5

Aim: Experiments On Packet Capture tool: Wireshark

Packet Sniffer:

* Sniffs messages being sent/ received from your Computer.
* Store & display contents of various protocol fields in m
* Passive program
  → never sends packet itself
  → no packets addressed to it
  → receives a copy of all packets

Packet Sniffer Structure Diagnostic Tools

* Tcpdump - eg tcpdump -enx host 10.129.41.2-w exe
* Wireshark - wireshark -r exe3.out.

Description:-

Wireshark:

Wireshark, a network analysis tool formerly known as Ethereal, captures packets in real time & display them in human-readable format. Wireshark includes filters, color coding & other features that you can use wireshark to analyze the traffic flow on your network.

What we can do with Wireshark:

* Capture network Traffic
* Decode Packet protocol using dissectors.
* Define filters - capture & display.
* Watch smart statistics
* Analyze problems

Capturing & Analysing Packets Using Wireshark Tool

1. Create Filter to display only TCP/UDP Packets, inspect the packets & provide the flow graph

Procedure:

* Select Local Area Connection in Wireshark.
* Go to capture option.
* Select Stop capture automatically after 100 packets
* Then click start capture.
* Search Tcp packets in Search Box
* To see flow graph click Statistics flow graph.
* Save packets

2. Create a filter to display only ARP packets & inspect the packets.

Procedure:

* Go to Capture option.
* Then click start Capture
* Search ARP packets in search bar & save it

3. Create a Filter to display only DNS Packets & provide flow graph.

Procedure:

* Go to capture option & select stop capture automatically after 100 packets & click start capture.
* Select DNS packets & click Statistics flow graph
* Save packets

4. Create a Filter to display only HTTP packets & inspect the packets.

Procedure:

* Select local Area Connection in Wireshark.
* Go to Capture Option.
* Select Stop Capture automatically after 100 packets

* Then Click Start Capture.
* Search HTTP Packets in Search Bar. & save it.

5. Create a filter to display only IP/ICMP Packets & inspect the Packets.

Procedure:
* Select Local Area Connection in Wireshark.
* Go to Capture option.
* Select Stop capture & click Start capture.
* Search Icmp/Ip packets in Search Bar.

6. Create a filter to Display only DHCP packets & Inspect the packets.

Procedure:
* Select Local Area Connection & Capture option.
* Click Start capture
* Search DHCP Packets & save it

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Ex: 6

Aim: Write a program to implement error detection and correction using Hamming Code concept. Make a test run to input data stream & verify error correction feature

Error correction at Data link layer:

#Sender

```
def calculate_parity_bits(data):
    m = len(data)
    r = 0
    while (2**r) < (m + r + 1):
        r += 1
    return r

def hamming_encode(data):
    r = calculate_parity_bits(data)
    total = len(data) + r
    code = ['0'] * (total + 1)

    j = 0
    for i in range(1, total + 1):
        if (i & (i - 1)) != 0:
            code[i] = data[j]
            j += 1

    for p in range(r):
        position = 2**p
        parity = 0
        for i in range(1, total + 1):
            if i & pos:
                parity ^= int(code[i])
        code[pos] = str(parity)
    return ''.join(code[1:])

with open("sender.txt", "r") as file:
```

```
text = file.read()
binary_data = ''.join(format(ord(char), '08b')
    for char in text)
print("Original Text:")
print(text)
print("In Binary Data:")
print(binary_data)
ham_data = ham_code(binary_data)
with open("channel.txt", "w") as file:
    file.write(ham_data)
print("In hamming code:")
print(hamming_data)
print("\nData Successfully sent to channel.txt")
```

2. Receiver

```
def hamming_decode(code):
    code = list(code)
    n = len(code)
    error_position = 0
    p = 1
    while p <= n:
        parity = 0
        for i in range(p, n+1):
            if i & p:
                parity ^= int(code[i-1])
        if parity != 0:
            error_position += p
        p *= 2
    if error_position != 0:
        print("Error detected at position:", error_position)
        code[error_position-1] = (
            '1' if code[error_position-1] == '0'
```

```
            else '0'
        )
    print ('Error corrected successfully.')
else:
    print ("No error detected.")
data = ''
for i in range (1, n+1):
    if (i & (i-1)) != 0:
        data += code[i-1]
return data
with open ("transmitted.txt", "r") as file:
    received_data = file.read().strip()
print("received Hamming Code:")
print (rec_data)
binary_data = ham_decode (rec_data)
print("\n Recovered Binary Data:")
print (binary_data)
text=""
for i in range(0, len(binary_data), 8):
    byte = binary_data [i:i+8]
    if len(byte) == 8:
        text += chr(int (byte, 2))
print("\n Final output :")
print (text)
```

O/P: sender.py

Enter text: HELLO

Binary data: 01001000010010101100011111

Hamming code: 010111000100001010100111001100111

Receiver.py

No error detected

Received text: HELLO

error detected at pos = 2-3

error corrected

Received text: HELLO

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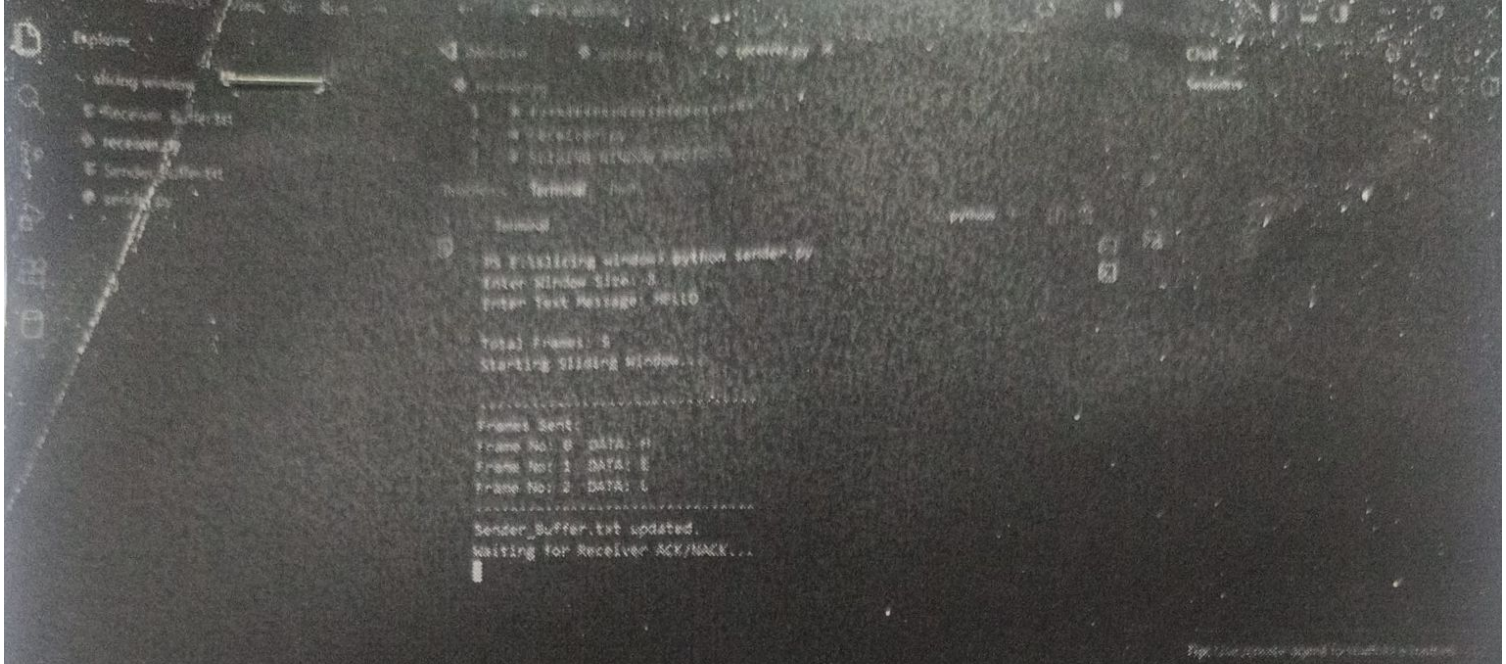
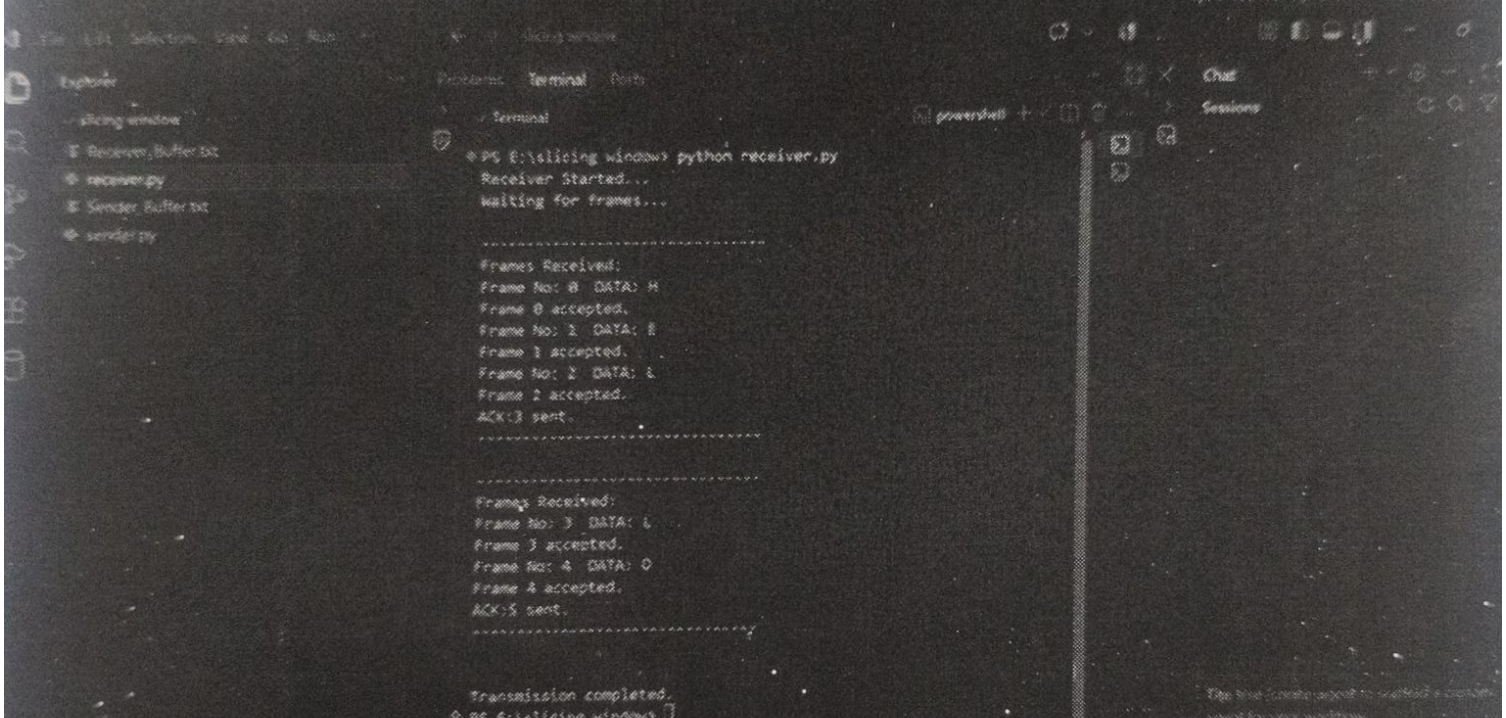
Ex:7

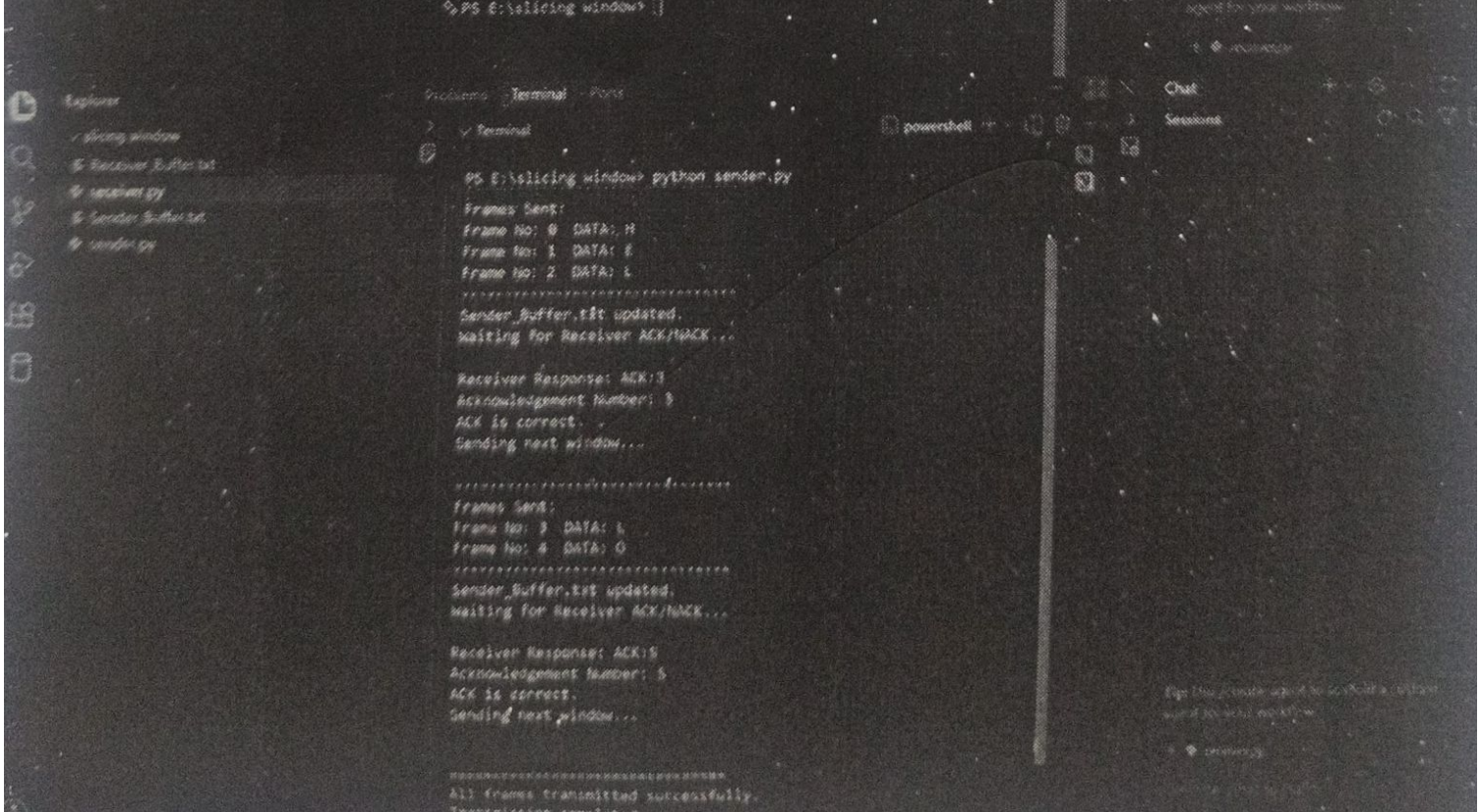
Aim: Write a program to implement flow control at data link layer using sliding window Protocol. Simulate the flow of frames from one node to another

Sender.py

```
import time
msg = input ("Enter msg: ")
w = int (input ("Enter window size: "))
start = 0
while start < len(msg):
    end = min (start + w, len (msg))
    with open ("Sender_Buffer", "w") as f:
        for i in range (start, end):
            f.write (f"{i} {msg [i]} \n")
    print ("\n Sending:")
    for i in range (start, end):
        print ("Frame", i, ":", msg [i])
    time.sleep(2)
    exec (open ("rec.py").read())
    with open ("Rec") as f:
        ack = int(f.read())
    print ("Ack:", ack)
    if ack > start:
        start = ack
print ("\n Msg Sent Successfully")
```


PS E:\slicing window> python sender.py
Enter Window Size: 3
Enter Text Message: HELLO
Total Frames: 5
Starting Sliding Window...
Frames Sent:
Frame No: 0 DATA: H
Frame No: 1 DATA: E
Frame No: 2 DATA: L
Sender_Buffer.txt updated.
Waiting for Receiver ACK/NACK...


PS E:\slicing window> python receiver.py
Receiver Started...
Waiting for frames...
Frames Received:
Frame No: 0 DATA: H
Frame 0 accepted.
Frame No: 1 DATA: E
Frame 1 accepted.
Frame No: 2 DATA: L
Frame 2 accepted.
ACK:3 sent.
Frames Received:
Frame No: 3 DATA: L
Frame 3 accepted.
Frame No: 4 DATA: O
Frame 4 accepted.
ACK:5 sent.
Transmission completed.
PS E:\slicing window>


PS E:\slicing window> python sender.py
Frames Sent:
Frame No: 0 DATA: H
Frame No: 1 DATA: E
Frame No: 2 DATA: L
Sender_Buffer.txt updated.
Waiting for Receiver ACK/NACK...
Receiver Response: ACK:3
Acknowledgement Number: 3
ACK is correct.
Sending next window...
Frames Sent:
Frame No: 3 DATA: L
Frame No: 4 DATA: O
Sender_Buffer.txt updated.
Waiting for Receiver ACK/NACK...
Receiver Response: ACK:5
Acknowledgement Number: 5
ACK is correct.
Sending next window...
All frames transmitted successfully.
Transmission completed.
PS E:\slicing window>

rec.py

```
with open("Send") as f:
    frames = f.readlines()
expected = 0
for line in frames:
    no, data = line.split()
    if int(no) == expected:
        print("Received:", data)
        expected += 1
    else: break
with open("Receiver_Buf", "w") as f:
    f.write(str(expected))
```

I/P:

Enter Msg: HELLO

Enter Window Size: 2

O/P:

Sending:

Frame 0 : H

Frame 1 : E

Ack : 2

Sending:

Frame 2 : L

Frame 3 : L

Ack : 4

Sending:

Frame 4 : O

Ack : 5

Message Sent Successfully

Result:

The program is executed successfully.

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Ex: 8

Aim:- To discover live Hosts using Nmap scans (ARP, ICMP/UDP) on the TryHackMe Platform Ro

Link: https://tryhackme.com/room/nmap01

Subnetworks:

A network segment is a group of computers connected through common medium, such as a switch or wi-fi access point.

3. Enumerating Targets:-

List: A list of IP addresses or Hostnames.

Range: A range of Ip address to scan

Subnet: A Complete subnet can be scanned using CIDR.

Discovering Live Hosts

ARP: Finds MAC address of a device on the local network.

ICMP: Uses ping request and replies to check host availabity.

Tcp: Sends packet to TCP ports to chec whether a host is reachable.

Udp: Send Udp packets to Identify reachab hosts.

Nmap Host Discovery Using ARP

ARP stands for Address Resolution

Protocol.

It sends requests to find the MAC address associated with an IP address. Active devices may reply to request

Cmd: nmap -PR -sn TARGETS

Nmap Host Discovery Using ICMP

ICMP stands for Internet Control Messag protocol.

It uses Echo Request & Echo Replies to check whether a Host is reachable.

Some firewalls may block ICMP packets

-PE - ICMP Echo Request

-PP - ICMP Timestamp Request

-PM - ICMP Address Mask Request

Cmd: nmap -sn -PE TARGETS

Nmap Host Discovery Using TCP & UPD

a) TCP SYN Ping:

Sends a TCP SYN packet to a Target POA.

A SYN/ACK or RST request can indicate that host is reachable

Ex: nmap -sn -PS23 TARGETS

5. TCP ACK Ping :-

Sends a TCP ACK packets & waits for an RST response.

It generally requires a privileged account.

Ex: nmap -sn -PA80 TARGETS.

6. UDP ping :-

Sends UDP packets to a target. A closed port may return an ICMP Port unreachable message, indicating that host is reachable.

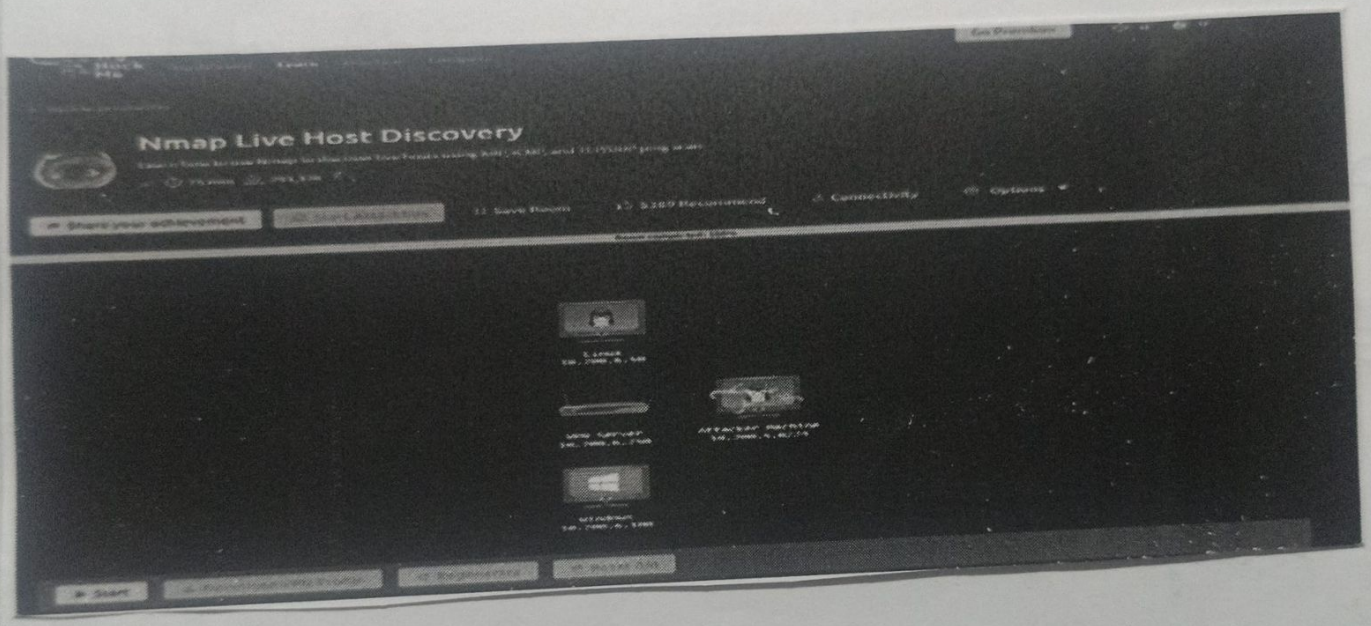
7. Reverse DNS Lookup

Reverse DNS lookup finds the hostname associated with an IP address.

-R - Forces reverse DNS lookups for all target IP addresses.

-n - Disables DNS resolution.

--dns - servers DNS_SERVER - Specifies a DNS server


Nmap Live Host Discovery

9/10/26

Ex: 8

(b)

Aim: To configure a Wireless LAN using Cisco Packet Tracer.

IP Configuration:

Device	IP Address	Subnet Mask
Router	192.168.0.1	255.255.255.0
Pc0	192.168.0.2	255.255.255.0
Pc1	192.168.0.3	255.255.255.0
Pc2	192.168.0.4	255.255.255.0

Procedure:

1. Open Cisco Packet Tracer & place One Linksys Wireless Router & three PCs.
2. Open the router & select the Administration tab. Set the Username & Password to admin & save the setting.
3. Select the Wireless tab & set the SSID to MotherNetwork.
4. Open Wireless Security, Select WEP as the Security mode, Set Key 1 to 0123456789.

Save the Settings.

5 Configure the router IP address as 192.168.0.1

6. Open each PC, select Desktop → IP Configuration → Static & enter the IP addresses given below

Connecting PCs to the Wireless Network

1. Open PC0 & select Desktop → PC → Wireless → Connect

2. Click Refresh & select the MotherNetwork Network.

3. Enter the WEP Security Key 0123456789 & click Connect.

4. Repeat the same steps for PC1 & PC2

5. Check the all three PCs are connected to the Wireless router.

6. Therefore, the Connection

is established.

Result:-

Therefore, the experiment is done successfully

9/10/26

Ex :- 9

Implementation of Subnetting in Cisco Packet Tracer.

Aim: To implement IP subnetting using Cisco Packet Tracer Simulator

Creating Network Topology.

Open Cisco packet Tracer & create a new network.

Add routers, Switches & PCs.

Connect the devices using suitable cables.

Assign IP addresses & subnet masks.

Subnetting:

The given network is 192.168.1.0/24.

A /27 subnet mask is used to create smaller network.

Subnet mask : 255.255.255.224

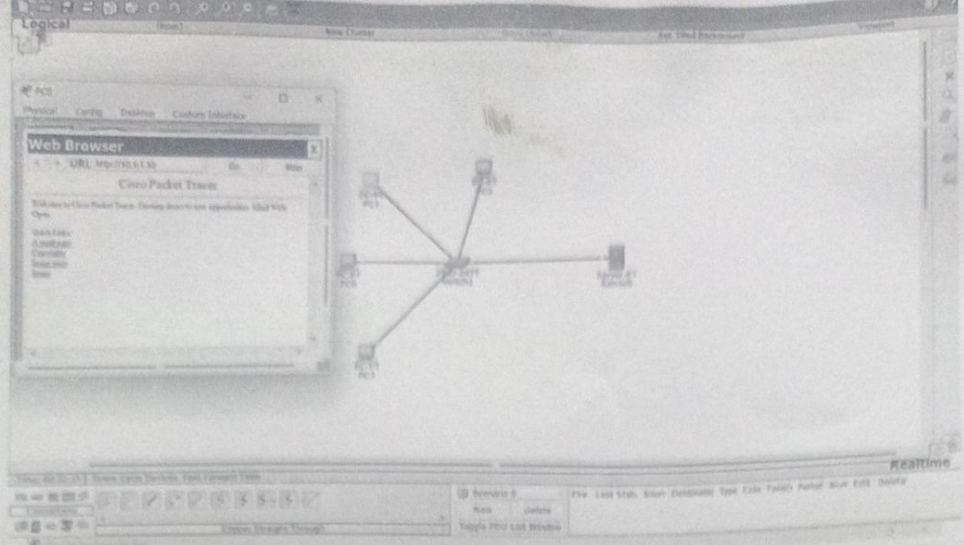
Total addresses per subnet : 32

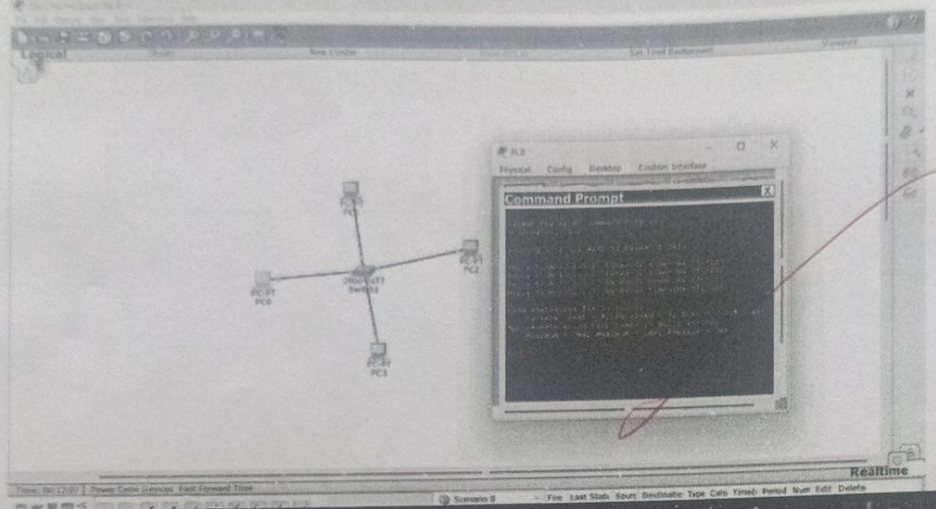
Usable host addresses : 30

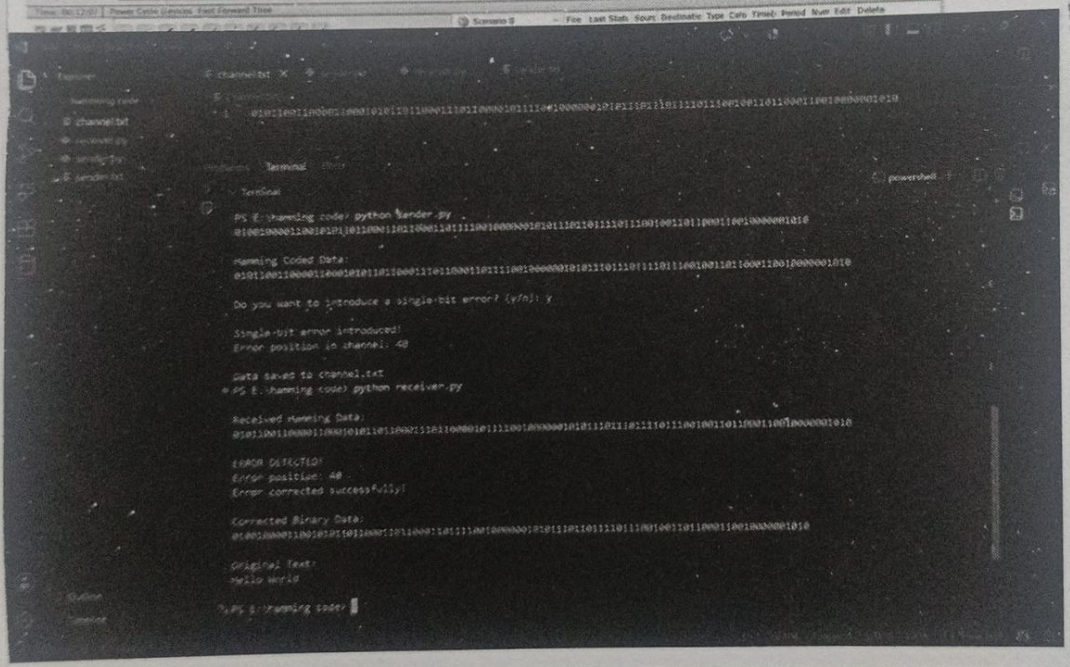
No. of subnet : 8

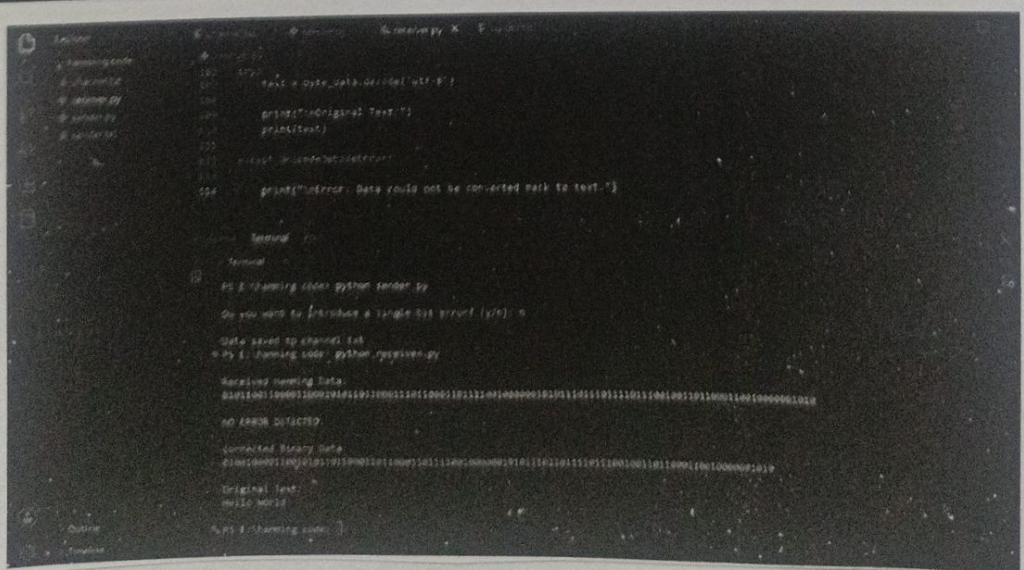
Router Configuration:

Enter these cmds in router CLI:


Logical
Web Browser
Cisco Packet Tracer
Realtime


Logical
Command Prompt
Realtime


Terminal
PS E:\hamming code> python sender.py
Hamming Coded Data:
Do you want to introduce a single-bit error? (y/n): y
Single-bit error introduced!
Error position in channel: 48
Data saved to channel.txt
PS E:\hamming code> python receiver.py
ERROR DETECTED!
Error position: 48
Error corrected successfully!
Original Text:
Hello World
PS E:\hamming code>


Terminal
PS E:\hamming code> python sender.py
Do you want to introduce a single-bit error? (y/n): n
Data saved to channel.txt
PS E:\hamming code> python receiver.py
NO ERROR DETECTED
Original Text:
Hello World

onfigure terminal
interface GigabitEthernet 0/0
ip address 192.168.1.1
255.255.255.224
no shutdown
exit.

Configure remaining interfaces with the respective Ip address.

Testing the Network :-

Assign Ip address, Subnet masks & default gateways to the Pcs.

Use the ping Cmd to test connectivity between devices.

Result :

The Implementation done Successfully.

9/10/26 Ex: 10

a) Interconnecting with Routers in Cisco packet Tracer.

Aim:- To design & Configure a simple internetwork Using a router in Cisco packet Tracer

Procedure:

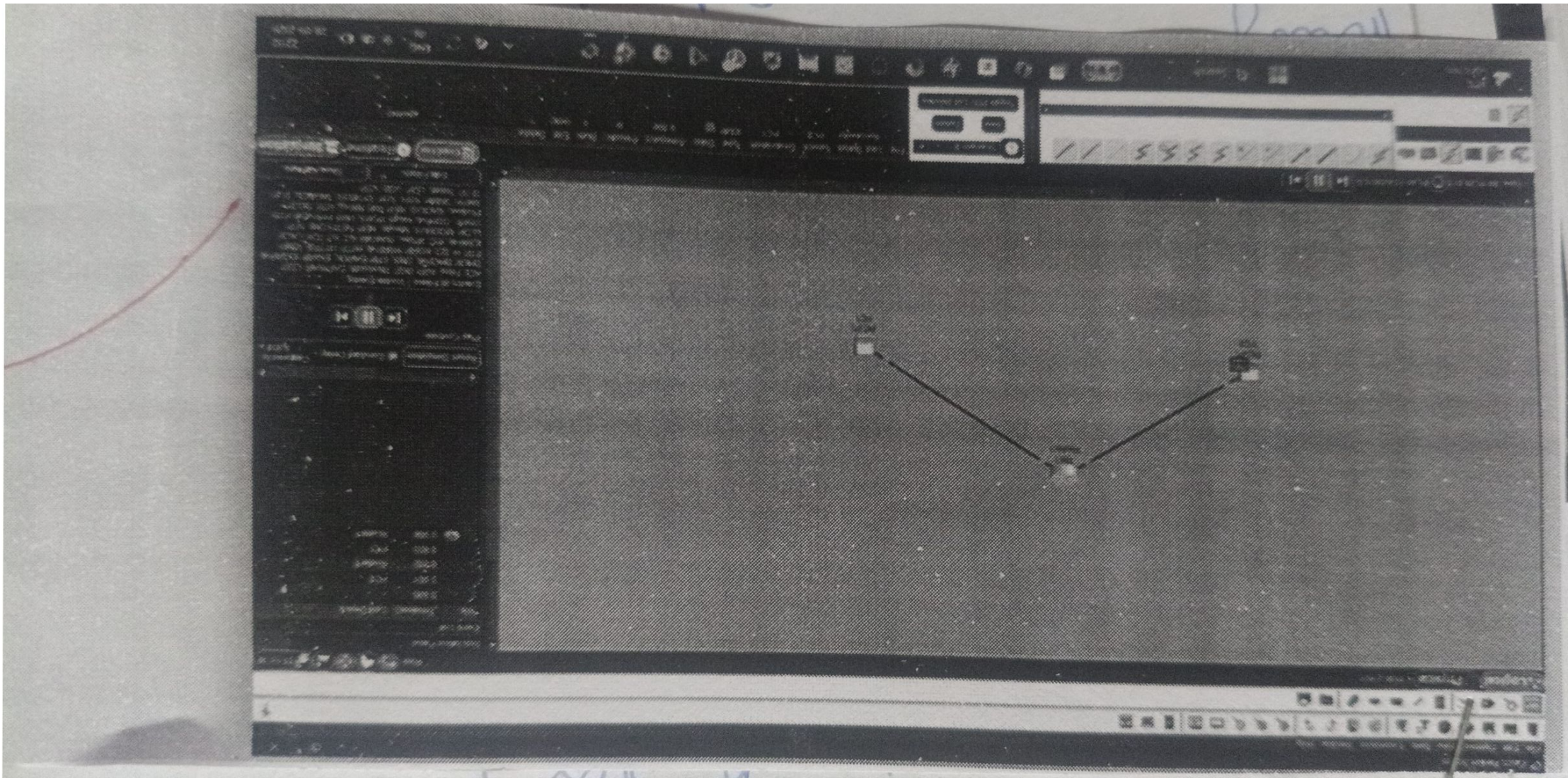
Step 1: Configure the Router.

1- Open the router CLI & press Enter.

2- Enter the following cmds:

enable
Configure terminal
interface FastEthernet 0/0
ip address 192.168.10.1
255.255.255.0
no shutdown
exit
Interface FastEthernet 0/1
ip address 192.168.20.1
255.255.255.0
no shutdown
exit

Step 2: Configure Pcs.



Open each PC & select Destop → IP configuration.

Assign the following IP addresses, Subnet masks, gateways.

Devices	IP address	Subnet mask
PC0	192.168.10.2	255.255.255.0
PC1	192.168.20.2	255.255.255.0

Step 3:- Connect the Devices.

1. Connect PC0 to Router1 FastEthernet 0/0 using a copper straight through cable.
2. Connect PC1 to Router1 FastEthernet 0/1 using copper straight through cable.
3. Wait until the interfaces become Active.

Step 4: Test Connectivity.

Result:

A simple internetwork was designed & tested.

b Ex: 10

Internetworking using Wireless Router, DHCP server & Internet Cloud

Aim: To design & configure an Internetwork using a wireless router, DHCP server, & Internet Cloud in Cisco Packet Tracer.

Procedure

Step 1: Build the network Topology.

1. Add a wireless router, cable modem, Internet Cloud, Cisco.com server, PC, & laptop.
2. Connect the PC to the router using a straight-through cable.
3. Connect the router to the cable modem, and the modem to the Internet Cloud.
4. Connect the Internet Cloud to the Cisco.com server.

Step 2: Configure the Wireless Router

1. Open the router GUI & set the SSID to Home Network.

2. Enable the DHCP server.
3. Set the DNS server address to 208.67.220.220.
4. Save the settings.

Step 3: Configure the laptop & PC.
1. Install the Wireless module in the laptop & connect it to Home Network
2. Select Desktop → IP configuration → DHCP on the pc.
3. Check that PC receives an Ip address.

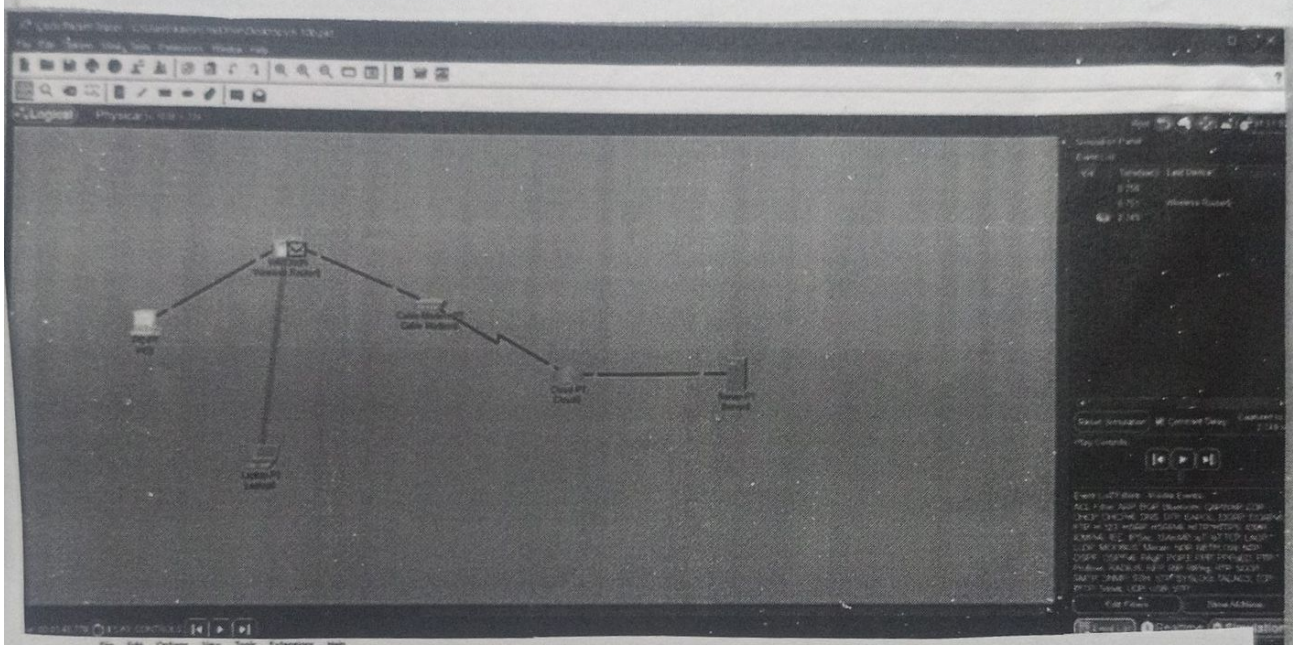
Step 4: Config Internet Cloud & Server
1. Config the Cloud with coaxial & Ethernet Connections.
2. Enable DHCP on Cisco.com Server.
3. Set Server Ip address to 208.67.220.220 & subnet mask to 255.255.255.0.
4. Enable DNS & add Cisco.com A record pointing to 208.67.220.220.

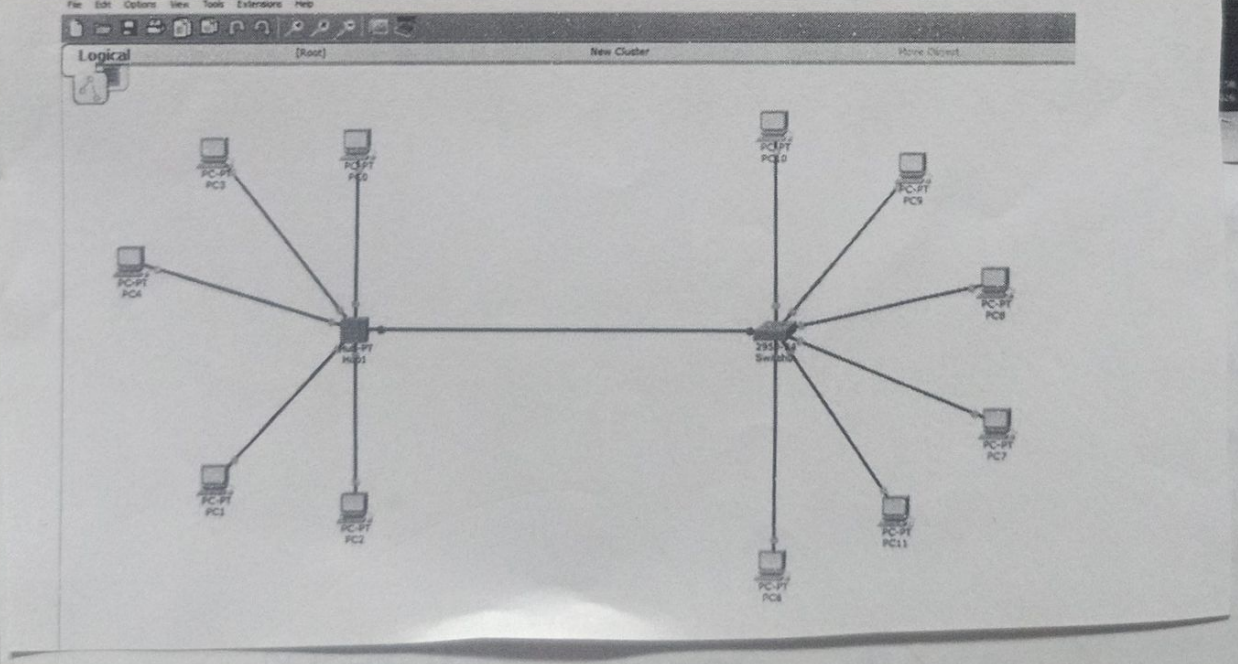
Steps: Test Connectivity
1. Open the PC Cmd Prompt

2- Run Ipconfig /release & Ipconfig /renew to refresh the IP config.

3- Use Ipconfig /all to verify the IP address & gateway.

4 Enter Ping cisco.com





Result:-

An Internetwork using a wireless router was tested.

21/10/26

Ex: 11

(a) Static Routing Configuration using Cisco Packet Tracer.

Aim: To simulate static routing config using Cisco Packet Tracer.

Procedure:

1. Open Cisco Packet Tracer & create a Topology with three routers & 2 pcs.
2. Connect the routers & configure their Ip addresses.
3. Open the router CLI & enter the cmds to config a static route

Enable
Configure Terminal
ip route 30.0.0.0 255.0.0.0
20.0.0.2
Exit
Show ip route static

4. Configure required routes on each router using the correct destination network & next hop Ip address.
5. To configure a backup route, assign

3. Verifying Static Routing

Use show ip route static to view static routes.

Use ping to test connectivity between networks.

Use tracert on a PC to identify the path taken by packets.

Disconnect main route to check whether backup route works.

Result.

Static routing was configured & verified.

9/10/25 Ex: 11

b) Simulate RIP Using Cisco Packet Tracer

Aim: To Simulate RIP (Routing Information Protocol) Using Cisco Packet Tracer.

Procedure:

1. Open Cisco Packet Tracer & connect three routers & two PCs as shown in the topology.
2. Assign IP addresses to the PCs. through Desktop → IP Configuration.
3. Configure the router interfaces Using the CLI

Ex: for Router()

enable
configure terminal
interface fastEthernet 0/1
ip address 10.0.0.1 255.0.0.0
no shutdown
exit

Configure Rip Routing

Enable Rip & advertise the directly connected networks on each Router.

Router():

router rip
Network 10.0.0.0
Network 192.168.1.0

Router 1:

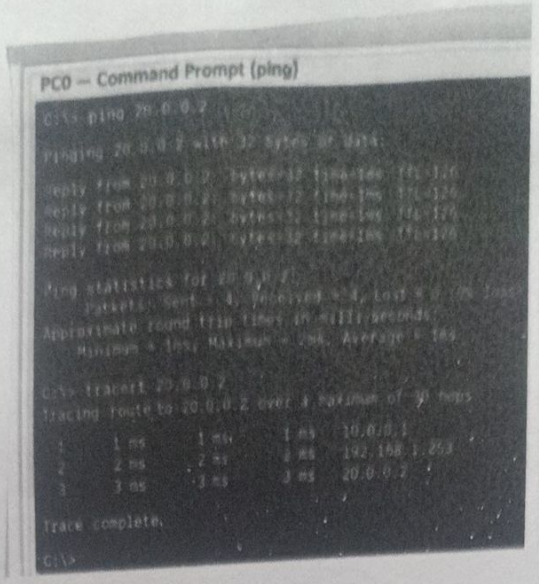
router rip
Network 192.168.1.0

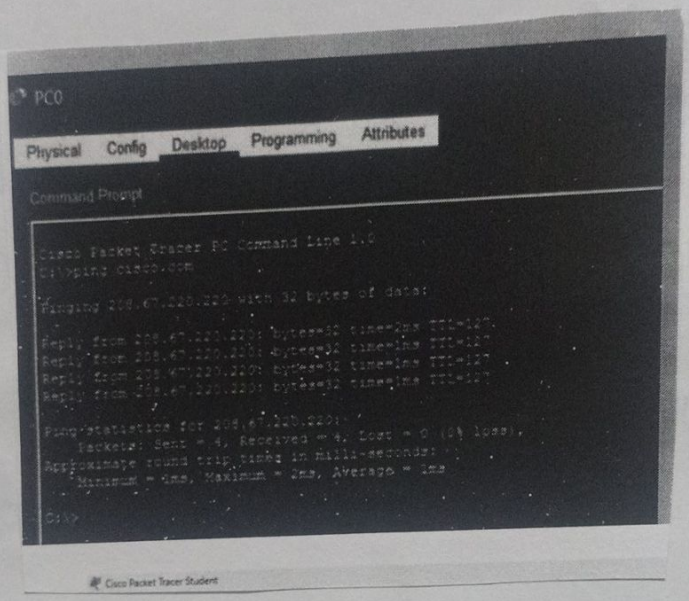
Router 2:

router rip
network 192.168.1.0
network 20.0.0.0

4. Verification

1. Open the Cmd on PC0 & enter ping 20.0.0.2 to test connectivity with PC1.
2. Use Tracert 20.0.0.2 to view the path taken by packets.
3. Disconnect the direct Serial link btw R0 & R2.
4. Run the ping & Traceroute Command Again
5. RIP should find the alternative route through R1 after the routing table update





Result:

RIP Routing was configured Successfully.

9/10/26 Ex:12

End-End Communication at Transport layer

a) Implement echo client server using Tcp/Udp sockets

b) Implement chat prog using socket

TCP

```
import socket
import threading
#--server--
def start_server():
    server = socket.socket(socket.AF_INET, socket.SOCK_
                           STREAM)
    server.bind('127.0.0.1', 5000)
    server.listen(1)
    print("[SERVER] Listening on 127.0.0.1:5000")
conn, addr = server.accept()
print(f"[SERVER] connected by {addr}")
while True:
    data = conn.recv(1024).decode()
    if not data or data.lower() == 'quit':
        print("[SERVER] Connection closed by
        client.")
        break
    print(f"[SERVER] Received: {data}")
    conn.send(f"Server received: [data]".encode())
conn.close()
server.close()
```

```
#--- Client ---
def start_client():
    client = socket.socket(socket.AF_INET,
        socket.SOCK_STREAM)
    print("[CLIENT] Connected to server")

    while True:
        msg = input("Enter msg (or 'quit'): ")
        client.send(msg.encode())
        if msg.lower() == 'quit':
            break
        response = client.recv(1024).decode()
        print(f"[SERVER RESPONSE] {response}")
    client.close()

#--- Main ---
if __name__ == "__main__":
    server_thread = threading.Thread(target=start_
        server, daemon=True)
    server_thread.start()
    start_client
```

O/P:-

```
Enter msg:
Hello
Receiver: hello
Enter msg:
Hi
Server received Hi
```

UDP

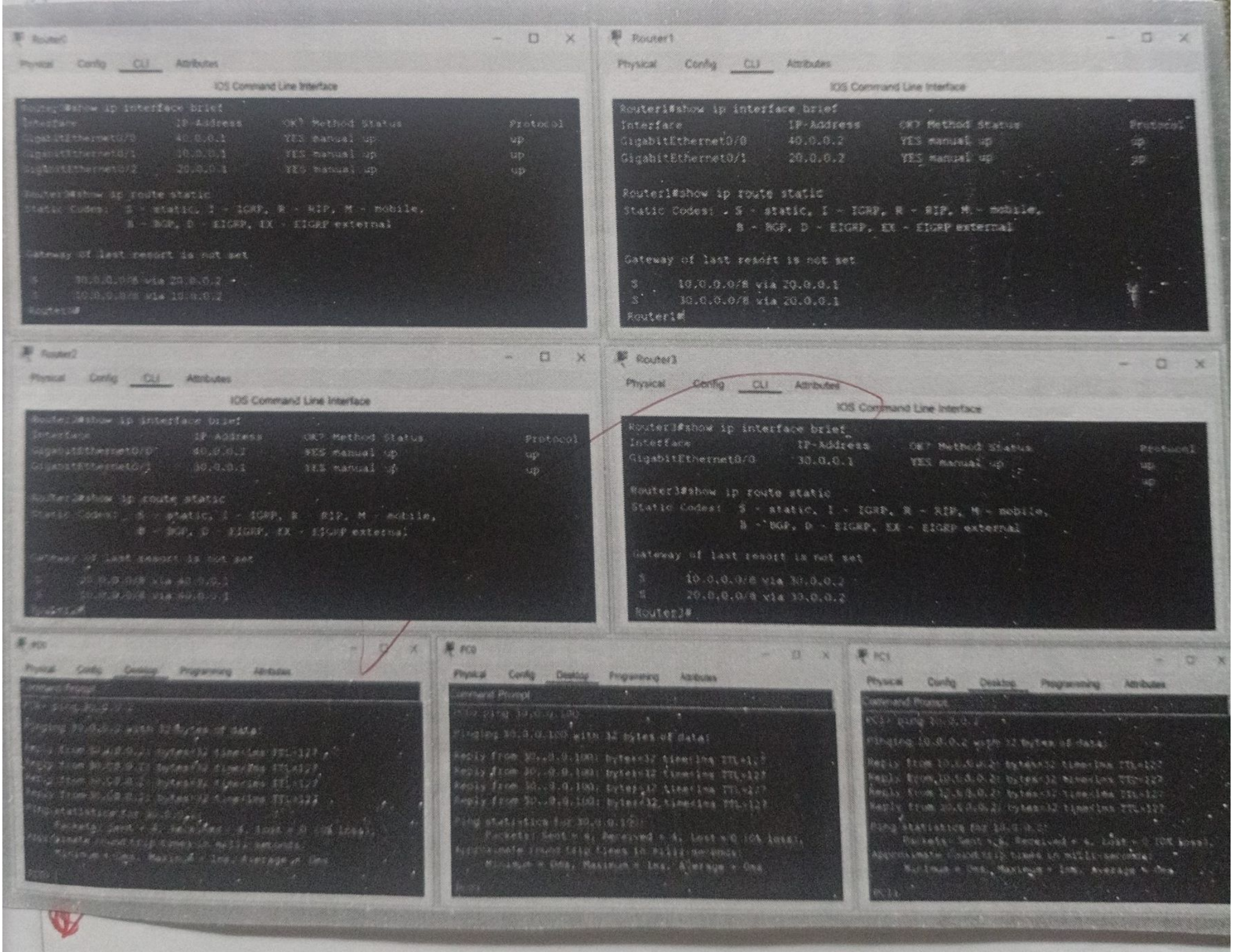
```
import socket
import threading
# --- server ---
def udp_server():
    s = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
    s.bind(('127.0.0.1', 12000))
    print("[SERVER] Running on 127.0.0.1:12000")
while True:
    data, addr = s.recvfrom(1024)
    msg = data.decode()
    if msg.lower() == quit
    break
    print(f"[SERVER] Receiver from {addr}: {msg}")
    s.sendto(data, addr) # Echo back
s.close()
# --- client ---
def udp_client():
    c = socket.socket(socket.AF_INET, socket.
        Sock_DGRAM)
    server = ('127.0.0.1', 12000)
while True:
    msg = input("Enter msg:")
    c.sendto(msg.encode(), server)
if msg.lower() == 'quit':
    print("[Client] closing connection...")
```

```
        break
print("[CLIENT] stopped.")

# ---- Main ----
if __name__ == "__main__":
    threading.Thread(target=udp_server,
    daemon=True).start()
    udp_client()
```

O/p:-

```
Enter msg: hi
Received: hi
Echo from Server: hi
```


Router1
Physical Config CLI Attributes
IOS Command Line Interface
Router1#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 40.0.0.2 YES manual up up
GigabitEthernet0/1 20.0.0.2 YES manual up up
Router1#show ip route static
Static Codes: S - static, I - IGRP, R - RIP, M - mobile,
B - BGP, D - EIGRP, EX - EIGRP external
Gateway of last resort is not set
S 10.0.0.0/8 via 20.0.0.1
S 30.0.0.0/8 via 20.0.0.1
Router1#
Router3
Physical Config CLI Attributes
IOS Command Line Interface
Router3#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 30.0.0.1 YES manual up up
Router3#show ip route static
Static Codes: S - static, I - IGRP, R - RIP, M - mobile,
B - BGP, D - EIGRP, EX - EIGRP external
Gateway of last resort is not set
S 10.0.0.0/8 via 30.0.0.2
S 20.0.0.0/8 via 30.0.0.2
Router3#
PC0
Physical Config Desktop Programming Attributes
PC2
Physical Config Desktop Programming Attributes
Pinging 30.0.0.100 with 32 bytes of data:
PC3
Physical Config Desktop Programming Attributes

9/10/25 Ex: 13

Implement your own Ping Program

```
import os
import platform
import subprocess
def ping(host="google.com", count=4):
param = "-n" if platform.system().lower() ==
    "windows" else "-c"
command = ["ping", param, str(count), host]
try:
    output = subprocess.check_output(cmd, universal_
        newlines=True)
    print(output)
except Exception as e:
    print(f"ping failed: {e}")
if __name__ == "__main__":
ping("8.8.8.8")
```

O/P:-

pinging 8.8.8.8 with 32 bytes of data.
Reply from 8.8.8.8: Bytes = 32 time = 6ms TTL=11
Reply from 8.8.8.8 bytes = 32 time = 9ms TTL=11

Ping Statistics for 8.8.8.8:
Packets: Sent = 4, Received = 4 Lost = 0
Minimum = 5ms Max = 9s, Avg = 4ms

9/10/25 Ex:-14

Write a code using RAW Sockets to implement packet sniffing

```
from scapy.all import sniff
from scapy.layers.inet import IP, TCP, UDP, ICMP
def packet_callback(packet):
if IP in packet:
    ip_layer = packet[IP]
    protocol = ip_layer.proto
    src_ip = ip_layer.src
    dst_ip = ip_layer.dst

Protocol_name = ""
if protocol == 1:
  Protocol_name = "ICMP"
elif protocol == 6:
    Protocol_name = "TCP"
elif protocol == 17:
    Protocol_name = "UDP"
else:
    Protocol_name = "Unknown Protocol"

print(f"Protocol: {protocol_name}")
print(f"Source Ip: {src_ip}")
print(f"Dest Ip: {dst_ip}")
print("-" * 50)
```

sniff(iface = "Wi-fi", prn = packet_callback, filter = "ip" store = 0)

O/P:-

protocol : TCP

Source - 57.158.141.117

Dest Ip : 192.168.1.6

protocol : UDP

Source Ip : 192.168.1.1

Dest Ip : 192.168.1.6

Completed

13/10/26